

Chapter 6 — Linear Model Selection and Regularization

Detailed Solutions

Solutions in L^AT_EX

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1 Exercise 6.2 (Conceptual)

Enunciation

Compare best subset selection, ridge regression, and lasso. Under what circumstances might one be preferred over the others?

Solution

- **Best Subset Selection:** Evaluates all subsets of predictors, chooses the best according to C_p , AIC, BIC, or adjusted R^2 .

- Pros: Very interpretable, can find the truly best set if p is small.
- Cons: Computationally infeasible for large p , prone to instability.

- **Ridge Regression:** Shrinks coefficients with an L_2 penalty:

$$\min_{\beta} \left\{ \sum (y_i - x_i^T \beta)^2 + \lambda \sum \beta_j^2 \right\}.$$

- Pros: Handles multicollinearity, reduces variance.
- Cons: Does not perform variable selection (all predictors remain).

- **Lasso:** Shrinks coefficients with an L_1 penalty:

$$\min_{\beta} \left\{ \sum (y_i - x_i^T \beta)^2 + \lambda \sum |\beta_j| \right\}.$$

- Pros: Produces sparse models (variable selection).
- Cons: Can be unstable when predictors are highly correlated.

Practical Choice:

- Small p : Best subset feasible and interpretable.
- Many correlated predictors: Ridge helps stabilize estimates.
- High-dimensional $p \gg n$: Lasso performs automatic selection.

2 Exercise 6.9 (Numerical)

Enunciation

Consider the effect of L_1 penalization (lasso) on regression coefficients. Provide an example showing how increasing λ changes coefficients.

Solution

Suppose OLS estimates for $y = \beta_1 x_1 + \beta_2 x_2$ are:

$$\hat{\beta}_1 = 2, \quad \hat{\beta}_2 = 1.5.$$

The lasso solves:

$$\min_{\beta_1, \beta_2} \left(RSS + \lambda(|\beta_1| + |\beta_2|) \right).$$

Effect of λ :

- $\lambda = 0$: Coefficients equal OLS: (2, 1.5).
- $\lambda = 1$: Estimates shrink, e.g. (1.7, 1.1).
- $\lambda = 3$: Further shrinkage, possibly (1.0, 0.0).

Geometric Intuition:

- Ridge penalty (L_2) has circular constraint $\sum \beta_j^2 \leq t$.
- Lasso penalty (L_1) has diamond-shaped constraint $\sum |\beta_j| \leq t$.
- Corners of the diamond encourage exact zeros, producing sparse solutions.

3 Exercise 6.12 (Python)

Enunciation

Apply LassoCV and RidgeCV to the *Hitters* dataset. Report selected tuning parameters and interpret differences in coefficient estimates.

Solution

We standardize predictors, apply cross-validation, and compare the results.

```
import pandas as pd
import numpy as np
from sklearn.linear_model import LassoCV, RidgeCV
from sklearn.preprocessing import StandardScaler
from sklearn.datasets import fetch_openml

hitters = fetch_openml(name="Hitters", version=1, as_frame=
    True)
X = pd.get_dummies(hitters.data, drop_first=True)
y = hitters.target.astype(float)

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

lasso = LassoCV(cv=10, random_state=42).fit(X_scaled, y)
print("Best lambda (Lasso):", lasso.alpha_)
print("Non-zero coefficients:", np.sum(lasso.coef_ != 0))

ridge = RidgeCV(alphas=np.logspace(-3,3,100), cv=10).fit(
    X_scaled, y)
print("Best lambda (Ridge):", ridge.alpha_)
```

Interpretation:

- Lasso sets some coefficients exactly to zero, selecting only a subset of predictors.
- Ridge keeps all predictors, but shrinks them.
- Choice depends on whether variable selection or stability is the main goal.